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Studierichting: Informatica
Oefeningenreeks 3A nr. 36.16

34.16) Bereken de afgeleide van de fucntie: $f(x) = \ln(x^3 + \sqrt{x^6 + 1})$

$$f'(x) = \frac{1}{x^3 + \sqrt{x^6+1}} \cdot (3x^2 + \frac{1}{2}(x^6+1)^{\frac{-1}{2} \cdot 6x^5})$$

$$f'(x) = \frac{1}{x^3 + \sqrt{x^6+1}} \cdot (3x^2 + \frac{6x^5}{2\sqrt{x^6+1}})$$

$$f'(x) = \frac{1}{x^3 + \sqrt{x^6+1}} \cdot \left(\frac{3x^2\sqrt{x^6+1} + 3x^5}{\sqrt{x^6+1}}\right)$$

$$f'(x) = \frac{3x^2(\sqrt{x^6+1} + x^3)}{(x^3 + \sqrt{x^6+1})(\sqrt{x^6+1})}$$

$$f'(x) = \frac{3x^2}{\sqrt{x^6+1}}$$

References